SQLiteStudio (3.4.21) - [weather_data (weather_data)]

Database Structure View Tools Help

Structure Data Constraints Indexes Triggers DDL

Grid view Form view

Total rows loaded: 5

	record_id	city_id	timestamp	temperatu	humidity	pressure h	wind spee	weather_cond	rain_1h	snow_1h
1	1	1	2026-03-17T15:38:39	25.05	50	1009	2.06	haze	NULL	NULL
2	2	2	2026-03-17T15:38:40	26.99	65	1010	4.12	haze	NULL	NULL
3	3	3	2026-03-17T15:38:40	21.84	72	1013	4.92	thunderstorm	NULL	NULL
4	4	4	2026-03-17T15:38:41	28.36	75	1011	4.63	few clouds	NULL	NULL
5	5	5	2026-03-17T15:38:42	25.97	83	1010	5.14	haze	NULL	NULL

SQLiteStudio (3.4.21) - [cities (weather_data)]

Database Structure View Tools Help

Structure Data Constraints Indexes Triggers DDL

Grid view Form view

Total rows loaded: 5

	city_id	city_name	country	latitude	longitude	created_at
1	1	Delhi	IN	NULL	NULL	2026-03-17 15:38:39
2	2	Mumbai	IN	NULL	NULL	2026-03-17 15:38:40
3	3	Bangalore	IN	NULL	NULL	2026-03-17 15:38:40
4	4	Chennai	IN	NULL	NULL	2026-03-17 15:38:41
5	5	Kolkata	IN	NULL	NULL	2026-03-17 15:38:42

SQLiteStudio (3.4.21) - [highest average temperature]

Database Structure View Tools Help

Databases

Filter by name

weather_data (SQLit

Tables (2)

cities

weather...

Views

weather_data

Query History

```
-- Which city has the highest average temperature? --
SELECT c.city_name,
       AVG(w.temperature_c) AS avg_temp
FROM weather_data w
JOIN cities c ON c.city_id = w.city_id
WHERE w.timestamp >= datetime('now', '-30 days')
GROUP BY c.city_id
ORDER BY avg_temp DESC
LIMIT 1;
```

Grid view Form view

1 Total rows loaded: 1

	city_name	avg_temp
1	Chennai	28.36

SQLiteStudio (3.4.21) - [temperature trends]

Database Structure View Tools Help

Databases

Filter by name

weather_data (SQLit

Tables (2)

cities

weather...

Views

weather_data

Query History

```
-- What are the temperature trends over the last 30 days? --
SELECT
    c.city_name,
    date(w.timestamp) AS obs_date,              -- YYYY-MM-DD
    AVG(w.temperature_c) AS avg_temp_c
FROM weather_data w
JOIN cities c ON c.city_id = w.city_id
WHERE w.timestamp >= datetime('now', '-30 days')   -- last 30 days
GROUP BY c.city_name, date(w.timestamp)
ORDER BY c.city_name, obs_date;
```

Grid view Form view

1 Total rows loaded: 5

	city_name	obs_date	avg_temp_c
1	Bangalore	2026-03-17	21.84
2	Chennai	2026-03-17	28.36
3	Delhi	2026-03-17	25.05
4	Kolkata	2026-03-17	25.97
5	Mumbai	2026-03-17	26.99

SQLiteStudio (3.4.21) - [humidity correlate with rainfall]

Database Structure View Tools Help

Databases

Filter by name

weather_data (SQLit

Tables (2)

cities

weather...

Views

weather_data

Query History

```sql
-- How does humidity correlate with rainfall? --
WITH stats AS (
    SELECT
        COUNT(*)                        AS n,
        SUM(humidity)                   AS sum_x,
        SUM(rain_1h)                    AS sum_y,
        SUM(humidity * humidity)        AS sum_x2,
        SUM(rain_1h * rain_1h)          AS sum_y2,
        SUM(humidity * rain_1h)         AS sum_xy
    FROM weather_data
    WHERE rain_1h IS NOT NULL           -- only rows with rain data
)
SELECT
    (n * sum_xy - sum_x * sum_y) /
    SQRT( (n * sum_x2 - sum_x * sum_x) * (n * sum_y2 - sum_y * sum_y) )
        AS pearson_corr
FROM stats;
```

Grid view Form view

Total rows loaded: 1

	pearson_corr
1	NULL

SQLiteStudio (3.4.21) - [most extreme weather conditions]

Database Structure View Tools Help

Databases

Filter by name

weather_data (SQLit

Tables (2)

cities

weather...

Views

weather_data

Query History

```sql
--Which seasons have the most extreme weather conditions? --
SELECT
    c.city_name,
    CASE
        WHEN strftime('%m', w.timestamp) IN ('12','01','02') THEN 'Winter'
        WHEN strftime('%m', w.timestamp) IN ('03','04','05') THEN 'Spring'
        WHEN strftime('%m', w.timestamp) IN ('06','07','08') THEN 'Summer'
        ELSE 'Autumn'
    END AS season,
    MIN(w.temperature_c) AS min_temp_c,
    MAX(w.temperature_c) AS max_temp_c
FROM weather_data w
JOIN cities c ON c.city_id = w.city_id
GROUP BY c.city_name, season
ORDER BY c.city_name,
    CASE season
        WHEN 'Winter' THEN 1
        WHEN 'Spring' THEN 2
        WHEN 'Summer' THEN 3
        WHEN 'Autumn' THEN 4
    END;
```

Grid view Form view

Total rows loaded: 5

	city_name	season	min_temp_c	max_temp_c
1	Bangalore	Spring	21.84	21.84
2	Chennai	Spring	28.36	28.36
3	Delhi	Spring	25.05	25.05
4	Kolkata	Spring	25.97	25.97
5	Mumbai	Spring	26.99	26.99

SQLiteStudio (3.4.21) - [peak temperature hours for each city]

Database Structure View Tools Help

Databases

Filter by name

weather_data (SQLit

Tables (2)

cities

weather...

Views

weather_data

Query History

```sql
-- What are the peak temperature hours for each city? --
WITH hourly_avg AS (
    SELECT
        c.city_name,
        strftime('%H', w.timestamp) AS hour_24,    -- HH in 24-h format
        AVG(w.temperature_c) AS avg_temp_c
    FROM weather_data w
    JOIN cities c ON c.city_id = w.city_id
    GROUP BY c.city_name, hour_24
),
ranked AS (
    SELECT
        city_name,
        hour_24,
        avg_temp_c,
        ROW_NUMBER() OVER (PARTITION BY city_name ORDER BY avg_temp_c DESC) AS rn
    FROM hourly_avg
)
SELECT city_name, hour_24 AS peak_hour, avg_temp_c AS peak_avg_temp
FROM ranked
WHERE rn = 1
ORDER BY city_name;
```

Grid view Form view

1 Total rows loaded: 5

	city_name	peak_hour	peak_avg_temp
1	Bangalore	15	21.84
2	Chennai	15	28.36
3	Delhi	15	25.05
4	Kolkata	15	25.97
5	Mumbai	15	26.99